

PROMPT ENGINEERING

MODEL USED : CLAUDE AI

1.Zero shot + Role based:

Subject: Are you getting the most out of your data?

Hi [First Name],

Most teams I talk to have the same problem: they're sitting on a goldmine of data but spending more time wrestling with dashboards than making decisions.

I'm reaching out because [Company] looks like a business that moves fast — and I'd hate for slow reporting to be the thing holding you back.

Our analytics platform helps teams like yours:

- Cut reporting time by up to 60%
- Spot trends before they become problems
- Share insights across teams without needing a data scientist in the room

[Customer Name] saw a 3x improvement in decision speed within 90 days of switching. Happy to share the full story if it's useful.

Would a 20-minute call this week or next make sense? No pitch deck — just a quick conversation to see if there's a fit.

[Your Name]

[Title] | [Company]

[Phone] | [Email]

PROMPT WITH ROLE:

You are a senior sales rep who has closed 200+ enterprise deals. You know that cold emails fail when they talk only about features. They work when they name a specific pain point, Write a cold email to the VP of Marketing about our analytics software.

Subject: Your campaign data is lying to you

Hi [First Name],

I've talked to dozens of VP of Marketing roles in the last year. Almost all of them have the same quiet problem:

By the time their team reports on what worked last quarter, it's too late to act on it.

They're making next month's budget calls based on data that's two weeks stale — from five different spreadsheets, stitched together by someone who should be doing strategy instead.

That lag is costing real money. Not in some abstract way — in mis-allocated spend, in campaigns that ran too long, in channels that looked fine on the dashboard but were bleeding quietly underneath.

[Your Software] closes that gap. Marketing leaders at [Relevant Company A] and [Relevant Company B] now see campaign performance in real time — unified, no manual pulls, no waiting on the data team.

[First Name], I'm not going to pitch you on features. I'd rather spend 20 minutes showing you exactly where your current reporting setup is leaving money on the table — and let you decide if it's worth fixing.

Are you open to a call this week?

[Your Name]

[Title] | [Company]

[Phone]

2.Few shot+ Structured:

[Check your sleep score →]

08:38

Claude responded: This copy is genuinely excellent — sharp, self-aware, and speaks directly to how the audience actually thinks.

This copy is genuinely excellent — sharp, self-aware, and speaks directly to how the audience actually thinks. Here's a quick breakdown of what's working and one thing to watch:

What's landing well:

- **"We watched you sleep. In a cool way."** — Disarms the obvious creepiness of the premise in four words. Smart.
- **"stop guessing why you feel like garbage at 2pm"** — This is the pain point in plain English. No jargon, no "optimize your recovery." Real.
- **"Judge yourself accordingly"** — Earns a laugh while implying the data is actually useful. Tone is perfect for a fitness audience.
- **"No setup needed"** — Buried but powerful. Removes the #1 objection before it forms.
- **"Fair warning — the data might hurt your feelings"** — Great closer. Creates anticipation and makes the CTA feel like a dare rather than a button.

One thing to consider:

The middle section (woke up, time to fall asleep, "8 hours was actually 5") is the most valuable part informationally, but it reads slightly list-like in a copy that otherwise flows conversationally. You could tighten it:

It also breaks down your real rest vs. the hours you spent staring at the ceiling — because "8 hours" and 8 hours are not the same thing.

Keeps the voice, cuts the clinical feel.

Overall: This doesn't read like a feature announcement. It reads like a friend texting you about something cool.

PROMPT 2:

Sale Announcement (End of Season) Subject: 30% off. No catch. Okay, one catch. The catch is it ends Sunday. Everything in the store — bands, chargers, straps, the works — 30% off. We'd say "while stocks last" but honestly we have plenty. We just want you to feel urgency. Is it working? If you've been thinking about upgrading your strap or grabbing a second charger for the office, this is the moment. If you haven't been thinking about it, maybe start. We're not going to beg. But we're also not running this sale again until we feel like it. Code: PULSE30 at checkout. Ends Sunday 11:59pm. [Shop the sale →]

This is confident, punchy copy with a distinct brand voice. Here's the breakdown:

What's working brilliantly:

- **"No catch. Okay, one catch."** — The self-correction in the subject line is a pattern interrupt. Most people open just to find out what the catch is. That's a win before they've read a word of the email.
- **"We just want you to feel urgency. Is it working?"** — Breaking the fourth wall on a marketing trope is bold and it works *because* the brand has earned enough goodwill to joke about it. It makes the reader feel in on it rather than sold to.
- **"We're not going to beg. But we're also not running this sale again until we feel like it."** — Casual dominance. It flips the power dynamic without being arrogant. The "until we feel like it" is doing a lot of heavy lifting.

One thing to watch:

"If you haven't been thinking about it, maybe start."

This line is trying to be dry and funny but it's the one place the voice slightly loses footing — it nudges the reader toward "why am I even on this list?" Consider replacing with something that redirects the energy:

If you haven't been thinking about it — a spare charger for the office is the kind of thing you'll thank yourself for in three months.

Keeps the casual tone but plants a specific, useful thought instead of a shrug.

One small structural note:

The code and deadline are slightly buried. Given that urgency is literally the point of the email, consider making **PULSE30** and **Sunday 11:59pm** visually impossible to miss — bold, their own line, maybe repeated near the CTA button. The copy earns the click; the mechanics should make it frictionless.

Overall: This has a real voice — rare in ecomm email. The brand sounds like a person, not a template. That's the hardest thing to get right, and it's already here.

CRAFT PROMPT:

Review Structure :

1. Executive Summary

- Overall code quality score
- Production readiness assessment
- Scalability assessment
- Maintainability assessment

2. Strengths

What is implemented well

- Good engineering decisions
- Reusable patterns
- Performance-conscious implementations

3. Issues & Improvement Areas

For each issue:

Issue

- What is wrong
- Why it matters Impact
- Performance
- Scalability
- Reliability
- Maintainability Recommendation
- Simpler explanation
- Best-practice approach Improved Code
- Refactored version of the specific section

MEETING TRANSCRIPT:

TRANSCRIPT BEGINS :

[00:00] Priya Sharma: Alright, let's get started. Nikhil's at the board meeting so he won't be joining, but he wants an update afterward, especially on the timeline. So let's make sure we're clear on where things stand. Arjun, do you want to kick off with the model update?

[00:32] Arjun Mehta: Sure. So the good news is the extraction model is performing significantly better after the retraining last week. We switched from the fine-tuned GPT-3.5 to a Mistral 7B base with LoRA adapters, and the field-level accuracy on invoices went from 78% to 91%. That's

on our internal test set of 2,200 documents. Priya Sharma: That's a big jump. Is that across all document types or just invoices? Arjun Mehta: Just invoices for now. Purchase orders are at about 84%, which is up from 72%, so also a solid improvement. But contracts are still a problem. We're hovering around 67% on the contract extraction, and honestly I'm not sure fine-tuning alone is going to get us where we need to be on those. [01:45] Rohit Desai: Yeah, the contract issue is structural.

Contracts have way more variability in layout — different firms, different jurisdictions, different formatting conventions. We've been talking about a two-stage approach where we first classify the contract type and then route it to a specialized extraction head, but that adds a lot of complexity to the pipeline. Priya Sharma: How much complexity are we talking?

Another two weeks? A month? Rohit Desai: Honestly, if we want to do it properly, probably three to four weeks of engineering time. And that's just the model work — Sneha's team would need to build out the routing

logic on the data pipeline side too. [02:48] Sneha Iyer: Hold on, let me push back on that a little. The routing logic isn't the hard part. We can have that up in a few days, it's basically a classifier output feeding into an Airflow branch operator. The hard part is the training data. Do we even have enough labeled contracts to train specialized extraction

heads? Last I checked we had maybe 400 labeled contracts total, and

that's across all types. Arjun Mehta: We have 430, to be exact. And if we

split by type — NDAs, service agreements, licensing agreements — the smallest bucket is licensing with only 67 examples. That's not enough to fine-tune anything reliable. Priya Sharma: So what's the alternative? We can't ship with 67% accuracy on contracts. That's the one document type our enterprise clients actually care about the most. [04:12] Arjun Mehta: I think the realistic options are: one, we invest in a labeling sprint and get the training data up to at least 200 per contract type before we attempt the two-stage approach. Two, we use a prompted approach for contracts instead of fine-tuning — basically long-context extraction with Claude or GPT-4 using carefully engineered prompts. Three, we descope contracts from the V1 launch and add them in V1.1. Kavya Nair: Option three is not going to fly with the sales team. I've already got two enterprise prospects — Reliance Digital and Tata Communications — who specifically asked about contract extraction in their last calls. If we pull that from V1, we might lose both deals Priya Sharma: Agreed, descopeing is a last resort. What about option two — the prompted approach? Arjun, is that viable? [05:30] Arjun Mehta: It's viable but it changes our cost structure. Right now the fine-tuned Mistral model runs on our own GPU cluster — inference cost is basically zero after the initial hardware investment. If we switch contracts to a prompted approach with an external API, we're looking at roughly 12 to 15 paisa per page, which adds up fast for contract-heavy clients. Rohit Desai: It's also

slower. Fine-tuned model does extraction in about 1.2 seconds per page.

A prompted approach with Claude is more like 6 to 8 seconds depending on contract length. For a 40-page contract, that's over five minutes.

Kavya Nair: Is five minutes a dealbreaker though? These aren't real-time use cases. A client uploads a contract and gets the extraction back in five minutes — that's still way faster than their current process, which is someone manually reading the thing for an hour. [06:52] Sneha Iyer: I'm less worried about the latency and more worried about the reliability. If we're calling an external API for a core feature, we need proper retry logic, fallback handling, rate limiting — all of that. And we need to figure out the data residency question. Some of these enterprise clients have strict requirements about where their data gets processed. Are we okay sending their contracts to Anthropic's API? Priya Sharma: That's a good point. Kavya, do we know the data residency requirements for Reliance and Tata? Kavya Nair: Reliance definitely has a data residency clause — everything has to stay within India. I'd need to check the specifics for Tata. Let me follow up with their procurement team and get clarity this week. Priya Sharma: Okay, please do that by Monday. We can't make a technical decision without knowing the constraints. [08:15] Priya Sharma: Alright, here's what I'm thinking. Let's pursue option one and two in parallel. Arjun, can you and Rohit start the labeling sprint for contracts? Get the annotation team to prioritize contract labeling — I

want at least 200 labeled examples per contract type within two weeks.

Arjun Mehta: That's aggressive. The annotation team is currently finishing the invoice validation batch. We'd need to pull them off that.

Priya Sharma: The invoice model is already at 91%. We can pause the invoice annotations. Contracts are the bottleneck. Arjun Mehta: Fair

enough. I'll talk to Meera on the annotation team today and get them redirected. [09:35] Priya Sharma: Good. And simultaneously, Rohit, can

you prototype the prompted approach? I want to see what accuracy we can get with a well-engineered prompt on the 430 contracts we already

have. Give me a benchmark by end of next week. Rohit Desai: I can do the benchmark, but end of next week is tight. I'm also supposed to be

fixing the table extraction bug that's been blocking the QA team. Sneha Iyer: That table extraction bug is actually causing issues on my side too.

The downstream ETL pipeline breaks every time it encounters a document with malformed table outputs. We've had 14 pipeline failures

this week alone because of it. Priya Sharma: Okay, how severe is the table bug? Is it a 'things crash' problem or a 'things are slightly wrong'

problem? Sneha Iyer: Things crash. The pipeline fails, the document gets sent to a dead letter queue, and we have to manually reprocess. It's

costing us about two hours of engineering time per day just babysitting it.

[11:20] Priya Sharma: That's not acceptable. Rohit, fix the table bug first.

That's priority one. Then start the contract prompt benchmark. Can you

have the bug fixed by Monday? Rohit Desai: I think so, yeah. I've already identified the root cause — it's a parsing issue with nested tables. The model outputs valid JSON but the parser doesn't handle the nesting correctly. Should be a day of work, maybe two. Priya Sharma: Good. Fix it by Monday, then start the contract benchmark. Get me results by the Friday after — March 27th. Rohit Desai: Got it. [12:30]

Kavya Nair: One more thing on the timeline. Our V1 target launch date is April 18th. That's five weeks from now. Are we still tracking for that?

Because if the contract approach takes four more weeks of engineering after the benchmark results come in, we're already past the launch date.

Arjun Mehta: Honestly? I think April 18th is at risk. Even if the prompted approach works, integrating it properly with fallback handling, monitoring, and cost tracking is at least two weeks of work. And that's after we know which approach we're going with. Priya Sharma: I hear you, but I'm not ready to move the date yet. Let's see where we are after the benchmark results on March 27th. If we need to adjust, we'll adjust then. But I don't want us to preemptively relax the timeline. Sometimes deadlines make magic happen.

Kavya Nair: Fair, but can we at least flag it to Nikhil? He's going to ask. Priya Sharma: I'll talk to Nikhil. Let me frame it properly — we're on track for invoices and purchase orders, contracts are the open question, and we'll have clarity by end of month. [14:45] Priya Sharma:

Okay, let's move to the data pipeline update. Sneha, where are we on

the ingestion service? Sneha Iyer: Good progress, actually. The async ingestion pipeline is fully built and deployed to staging. We're processing about 1,200 documents per hour in testing, which is well above our target of 800. The S3-to-processing queue is stable, Airflow orchestration is clean, and we've got proper dead letter queues for failed documents. Sneha Iyer: The one outstanding issue — apart from the table bug that Rohit mentioned — is the OCR pre-processing for scanned documents. We're using Tesseract right now, and the quality is... not great on low-resolution scans. About 22% of our test documents are scanned, and for those, the extraction accuracy drops by 15 to 20 points because the OCR introduces errors before the model even sees the text. [16:30] Arjun Mehta: Have you looked at Azure Document Intelligence? Their OCR is significantly better than Tesseract, especially on degraded scans. We used it at my last company and it was night and day. Sneha Iyer: I've looked at it. The accuracy is better, no question. But it's another external API dependency, another cost line, and another data residency conversation. Kavya Nair: Also, sorry to keep bringing this up, but Azure has an India region, right? That might actually solve the data residency issue for both the OCR and the contract extraction if we go with a prompted approach through Azure OpenAI. Arjun Mehta: Azure OpenAI doesn't have the latest Claude models though. If we're going prompted, Claude is performing noticeably better than GPT-4 on

our extraction benchmarks. Like, 6 to 8 points better on contracts specifically. [18:10] Priya Sharma: Okay, we're going in circles a bit. Let me try to untangle this. Sneha, can you run a quick comparison of Tesseract versus Azure Document Intelligence this week? Use 50 of the worst scanned documents. I just need to know if the accuracy improvement justifies adding another vendor dependency. Sneha Iyer: Yeah, I can have that done by Wednesday. Priya Sharma: Great. And the data residency question — Kavya is going to find out the requirements from Reliance and Tata by Monday. Once we have that, a lot of these API decisions become clearer. Let's not go back and forth on Azure versus Anthropic until we actually know what's allowed. Kavya Nair: Makes sense. [19:45] Priya Sharma: Anything else before we wrap? We're almost at time. Rohit Desai: One quick thing — the GPU cluster is at 87% utilization. When we scale the annotation team and start running more training jobs for the contract models, we're going to hit capacity. Should I file a request for additional GPU allocation now, or wait? Arjun Mehta: File it now. The procurement process takes two weeks minimum. Better to have it in the queue and cancel than to need it and not have it. Priya Sharma: Agreed. Rohit, submit the GPU request today. Estimate what we'd need for the contract training runs and add a 30% buffer. Rohit Desai: Will do. [20:55] Kavya Nair: Oh wait, one more thing. The product demo for the Reliance team is scheduled for April 3rd.

That's three weeks out. What are we showing them? Because if contract extraction isn't ready, we need to frame the demo carefully. Priya

Sharma: Good catch. Let's show invoices and purchase orders — those are solid. For contracts, we can show a "coming soon" preview or a limited demo with NDAs only, since those are our strongest contract subtype. Kavya, can you draft a demo flow by end of next week and

share it with the team? Kavya Nair: Sure, I'll put together a demo script and share it by Friday the 20th. [22:10] Priya Sharma: Alright. Let me do

a quick recap. Arjun — redirect the annotation team to contract labeling, target 200 per type in two weeks. Rohit — fix the table extraction bug by

Monday, then start the prompted contract benchmark, results by March

27th. Also submit the GPU request today. Sneha — run the Tesseract

versus Azure OCR comparison by Wednesday. Kavya — get data

residency requirements from Reliance and Tata by Monday, and draft the demo flow by the 20th. I'll update Nikhil on the timeline situation.

Everyone clear? Arjun Mehta: Clear. Sneha Iyer: Got it. Rohit Desai:

Yep. Kavya Nair: All good. Priya Sharma: Great. Good progress on the

model side — 91% on invoices is a real milestone. Let's keep that

momentum going. Talk next week. TRANSCRIPT ENDS — 22:48

MEETING SUMMARY:

Weekly AI Pipeline Sync — Meeting Summary Date: March 12, 2026

Duration: 47 Minutes Attendees: Priya Sharma, Arjun Mehta, Sneha Iyer, Rohit Desai, Kavya Nair Absent: Nikhil Rao

Executive Summary

The team reviewed progress on the Document Intelligence Platform ahead of the planned April 18 V1 launch. Key highlights:

- Invoice extraction accuracy improved significantly from 78% to 91% after migrating to Mistral 7B + LoRA adapters.
- Purchase order extraction improved from 72% to 84%.
- Contract extraction remains the primary risk area at 67% accuracy.
- Async ingestion pipeline exceeded performance targets, processing 1,200 documents/hour versus the target of 800 documents/hour.
- A critical table extraction bug is causing pipeline failures and requires immediate attention.
- OCR quality for scanned documents remains a challenge.
- GPU cluster utilization has reached 87%, creating potential capacity constraints for future model training

Model Performance Update

Contract Extraction Options Discussed Option

1: Labeling Sprint + Two-Stage Architecture Approach:

1. Classify contract type.
2. Route to specialized extraction model.

Pros:

- Better long-term accuracy.

- Fully owned infrastructure.

Cons:

- Requires significant additional labeled data.
- Estimated 3–4 weeks of engineering effort.

Option 2: Prompted Extraction

Approach:

- Use Claude or GPT-4 with prompt engineering.

Pros:

- Faster path to improved accuracy.
- No immediate need for large training datasets.

Cons:

- Increased inference costs.
- Higher latency. • Data residency concerns.

Option 3: Remove Contracts from V1 Outcome:

- Rejected as a last-resort option due to enterprise customer demand.

Action Items:

Owner Action Item	Due Date
Arjun Redirect annotation team to contract labeling	Immediate

Arjun Reach 200 labeled examples per contract type	Within 2 weeks
Rohit Fix table extraction bug	Monday
Rohit Run prompted contract benchmark	March 27
Rohit Submit GPU capacity request	Today
Kavya Confirm data residency requirements]	Monday

Key Risks

1. Contract extraction accuracy remains below acceptable levels.
2. Limited labeled contract dataset.
3. Data residency constraints may restrict vendor choices.
4. OCR quality issues affect scanned document accuracy.
5. GPU capacity nearing limits.
6. April 18 launch date is at risk if contract solution requires significant additional engineering.